

YOUSSEF KASSAB

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PROFESSIONAL SUMMARY

Junior Software Developer and Tech Founder with hands-on experience in Full-Stack development, and building systems and software solutions. Proven to have the ability to design, build, deploy, and maintain production-ready systems under extreme constraints, including zero-connectivity environments, old or limited hardware.

PROFESSIONAL EXPERIENCE

YK Software Studio | Founder & Software Developer *Jun 2025 – Present*

- Designed and built a custom Admin Dashboard for the Degla Smart School website project, featuring a friendly UI/UX design that allows non-technical staff to manage the system independently with ease and proficiency.
- Developed a secure Node.js, Express, and MySQL backend system, protecting it from SQL injection threats by adding strict input sanitization and rate-limiting to ensure production-level reliability.
- Built the entire system using reliable open-source and custom tools, completely bypassing expensive subscriptions to cut long-term costs without losing efficiency.

HEMEX | Infrastructure Specialist & Instructor *2025*

- Designed and deployed a centralized OS provisioning pipeline across 30 workstations, utilizing lightweight Windows images pre-configured with development environments and Veyon lab management software.
- Reduced system boot times by 65% and increased overall performance, while enabling real-time monitoring and administrative control over 10+ concurrent clients across a localized network.

ENGINEERING & SYSTEMS ARCHITECTURE PROJECTS

Innovation Day (Man Ahyaha Foundation) | *Primary Developer & Keynote Speaker* | *Aug 2025*

- Volunteered as organizer and instructor in the youth program at Man Ahyaha.
- Deployed a local network using Docker for hosting Rocket.Chat and a custom scoring system for more than 100 users without internet access.
- Prepped and configured physical technical assets, including Arduino-based robotics and RC cars, for interactive student demonstrations.
- Maintained 100% uptime during the event by organizing a tech support team of 2 engineers and 4 technical instructors in hard environment.
- Delivered a live lecture addressing critical and logical thinking, problem-solving, soft skills, and mindset for kids and a non-technical audience.

Digitopia (3Im Quest) | *Project Manager & Backend Architect* | *Aug 2025 – Oct 2025*

- Developed a RESTful API with 36 endpoints and a relational database with 15 tables in Node.js and Express in 110+ files to manage and monitor real-time application states, user progression, AI states and usage.
- Designed and integrated systematic A/B testing protocols for an LLM educational evaluation pipeline to reduce AI hallucinations and maintain high accuracy.

- Stepped in as Project Manager to resolve team friction, organizing the workflow and distributing roles based on technical capabilities to successfully deliver the project and advance to the semi-finals.

HomeLab Infrastructure | *Systems Architect* | Nov 2025 – Present

- Developed and implemented a local integrated ecosystem using VMware on Windows 11 to serve multiple services and tools.
- Set up Home Assistant for local automation and system monitoring and configured AdGuard Home for network-wide DNS sinkholing and ad-blocking.
- Configured secure SMB sharing protocols for structured internal data management and media routing across the local network.

EMBEDDED SYSTEMS & HARDWARE RESEARCH

Independent Hardware Optimization | *Independent Researcher* | 2022- Present

- Repurposed end-of-life proprietary hardware by executing firmware-level modifications, extending device lifecycles, and reducing electronic waste.
- Safely bypassed OEM restrictions on an iPad Air (1st Gen) to restore administrative system access and deploy custom, lightweight operating environments.
- Optimized legacy gaming hardware for custom homebrew execution, deploying Open PS2 Loader (OPL) via Matrix on PlayStation 2 infrastructure, and configuring the Aurora dashboard and XEXMenu on Xbox 360 systems.

TECHNICAL SKILLS

- **Backend & APIs:** Node.js, Express.js, RESTful API Architecture, JavaScript (ES6+)
- **Data Engineering:** MySQL, Relational Database Design, Schema Optimization, Python
- **Infrastructure & DevOps:** Docker, VMware Hypervisor, LAN Architecture, OS Image Provisioning, Veyon Lab Management
- **Firmware & Hardware:** Embedded Systems, Arduino, Firmware Modification, AdGuard Home, Rate-Limiting, Payload Sanitization & Web Security

EDUCATION & ACHIEVEMENTS

Thanaweya Amma (Secondary Education) | *Cairo, Egypt* | *Expected 2026*

Curriculum: Ilmi Ryada — Advanced Mathematics & Science Stream

Academic Standing: Achieved Final Grade in Sec 2 of 87%

Athletics: Former National Kung-Fu Champion and long-term competitive swimmer, with a consistent record of high-pressure performance and sustained physical discipline.